

2023 AMC 10B Problem 11 - Solution

Review the full statement and step-by-step solution for 2023 AMC 10B Problem 11. Great practice for AMC 10, AMC 12, AIME, and other math contests

2023 AMC 10B Problem 11

**Problem:**

Suzanne went to the bank and withdrew \$800. The teller gave her this amount using \$20 bills, \$50 bills, and \$100 bills, with at least one of each denomination. How many different collections of bills could Suzanne have received?

Answer Choices:

- A. 45
- B. 21
- C. 36
- D. 28
- E. 32

Solution:

Note that the amount of money in \$20 bills must be a positive multiple of \$100, and the same is true for the \$50 bills and the \$100 bills. Then the amount of money apart from the \$20 bills must be \$200, \$300, \$400, \$500, \$600, or \$700. Thus the \$50 and \$100 bills must total to $n \cdot \$100$ for $2 \leq n \leq 7$. For each of these cases, the number of \$100 bills could be any integer from 1 to $n - 1$, with the remainder consisting of \$50 bills. Therefore the number of possibilities is

$$\sum_{n=1}^7 (n - 1) = 0 + 1 + \cdots + 6 = (B) \boxed{21}.$$

OR

Let a , b , and c represent the number of \$100 amounts made up of \$20 bills, \$50 bills, and \$100 bills, respectively. Then $a + b + c = 8$ and each of a , b , and c is an integer greater than or equal to 1. To count the number of positive integer solutions to this equation, consider 8 "stars" (indicated by asterisks) placed in a row, with 7 spaces (indicated by dashes) between pairs of consecutive stars:

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Now choose 2 of the 7 spaces between the stars in $\binom{7}{2} = 21$ ways and place vertical bars in those chosen spaces. This partitions the stars into three disjoint pieces, the integers a , b , and c . For example, placing a bar between the second and third stars and also the seventh and eighth stars

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gives the solution $a = 2$, $b = 5$, and $c = 1$. Thus, there are (B) 21 solutions to the equation.

Note: The argument used to count the number of integer solutions in the second solution is called a "stars-and-bars" or "sticks-and-stones" argument.

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